

# MISFIT: A Self-Supervised Pretraining Toolkit for 3D Medical Imaging

## A Preliminary Test Across Three BraTS 2026 Segmentation Tasks

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### Introduction

Since its 2015 introduction, U-Net has been cited over 100,000 times, and frameworks like nnU-Net remain the reproducible standard on BraTS and the Medical Segmentation Decathlon. Despite continued architectural innovation — transformers, boundary-aware losses — reported gains over nnU-Net are not always reproducible.

We address this with **MIST**, our modular, end-to-end segmentation framework. Using MIST, we placed **3rd** in both the 2024 BraTS Adult Glioma Post-Treatment challenge and the 2025 BraTS Pre-/Post-Treatment MRI challenge. For BraTS 2026 we enter the three tasks above, each testing a different kind of generalization.

#### This report asks two questions

- 01** System-level: does a **MISFIT**-pretrained SwinUNETR-V2 pipeline beat our established pocket nnU-Net baseline?
- 02** Isolated effect: holding architecture, patch size, and budget fixed, does MAE pretraining itself help over random initialization?

### Methods

| Table 1 · Dataset sizes and regions evaluated per task |                  |       |          |                    |
|--------------------------------------------------------|------------------|-------|----------|--------------------|
| TASK                                                   | COHORT           | TRAIN | HELD-OUT | REGIONS            |
| 1                                                      | Brain metastases | 1,196 | 179      | NETC, SNFH, ET, RC |
| 2                                                      | Pediatric tumors | 294   | 91       | ET, NET, CC, ED    |
| 3                                                      | BraTS-GoAT       | 1,351 | 450      | ET, TC, WT         |

#### Self-supervised pretraining via MISFIT

**MISFIT** is our in-development framework for pretraining 3D medical imaging encoders with masked autoencoding (MAE), independent of MIST. Using `misfit_index`, we pooled every training-split MRI volume across all three tasks — including Task 3's 1,137-patient unlabeled cohort — into **16,315** single-channel volumes (T1N/T1C/T2W/T2F indexed separately). A `swinunetr`-base encoder (18.6M params) was trained on 96<sup>3</sup> crops with 75% of the volume masked in non-overlapping 16<sup>3</sup> cubes, reconstructed via a lightweight transposed-convolution decoder under a per-patch variance-normalized MSE loss (AdamW, lr 1×10<sup>-4</sup>, 200 epochs, 4×A100). The pretrained encoder was then transferred into MIST's SwinUNETR-base segmentation architecture.

#### Three network arms

- Pocket nnU-Net (baseline)** — constant-width (32-filter) conv encoder-decoder, 128<sup>3</sup> patch, random init; our returning baseline from prior BraTS challenges.
- SwinUNETR-V2, MISFIT-pretrained** — same architecture, 96<sup>3</sup> patch, encoder initialized from MISFIT weights.
- SwinUNETR-V2, from scratch** — identical to the arm above except for random encoder initialization — the matched control.

The two SwinUNETR arms share architecture, patch size, and a 1,000-epoch fine-tuning budget, differing only in encoder initialization — isolating the effect of pretraining. The baseline differs from both in architecture, patch size, learning rate, and budget, so it measures a full pipeline-level contrast instead.

#### Training & evaluation

5-fold cross-validation, Dice + cross-entropy loss, AdamW, cosine schedule with 20-epoch warmup, bfloat16, GPU-resident DALI augmentation. The baseline trained up to 5,000 epochs/fold (mean 3,107–4,252 actual); both SwinUNETR arms trained exactly 1,000 epochs/fold. Inference used sliding-window Gaussian blending, 8-way flip test-time augmentation, and 5-fold averaging. Arms are compared with MIST's `mist_rank` — a BraTS-style per-patient average rank — and a two-sided Wilcoxon signed-rank test ( $\alpha = 0.05$ ).

### Discussion & Conclusion

*Held-out validation does not support the pooled MISFIT pipeline beating our established baseline — but the matched control shows MAE pretraining genuinely helps once architecture, patch size, and budget are held fixed.*

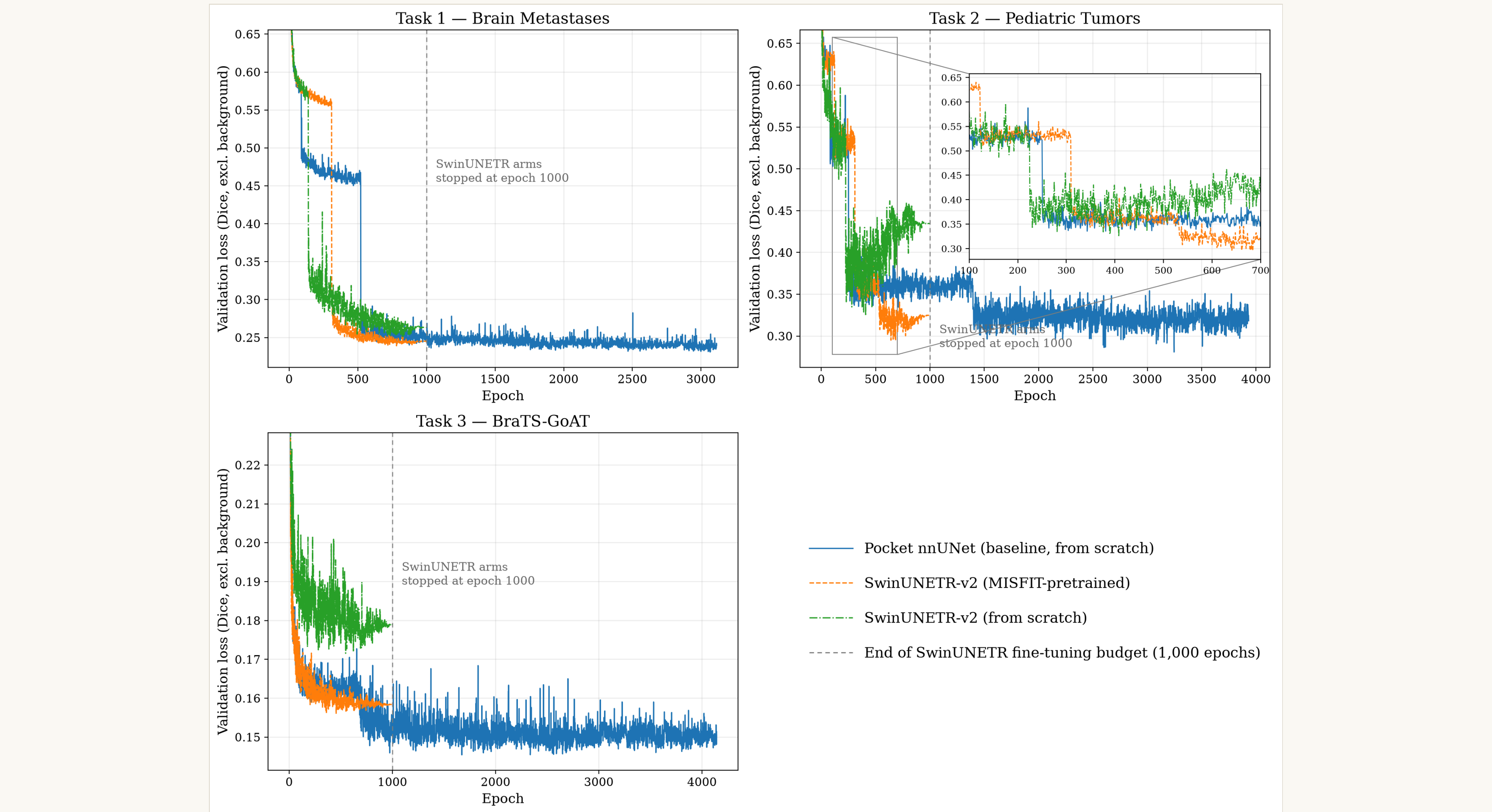
- Task 1's reversal is explainable.** Task 1 is scored lesion-wise, where a small boundary deficit invisible in per-voxel CV Dice is heavily amplified; a 1,000-epoch fine-tuning budget likely wasn't enough to refine small-lesion boundaries.
- Where pretraining helps most.** Task 2 enhancing tumor Dice improves from 0.518 → 0.670 under cross-validation; on Task 1 held-out, the resection cavity leads on every metric (Dice 0.886 → 0.923) while ET/TC/WT show no advantage — a concentrated, not uniform, gain.
- Transformers are data-hungry without pretraining.** Task 2 (294 cases) is the only task where the from-scratch SwinUNETR fails to converge — and the task where MISFIT initialization helps most.
- Final submission: pocket nnU-Net for all three tasks** — required by data-usage rules barring cross-task pooling, and independently supported by the held-out results themselves.
- This is a preliminary MAE schedule** — 200 epochs against the 800–1,600 convention. Longer pretraining and larger fine-tuning budgets are needed to test whether pretrained transformers can surpass fully converged CNNs on these tasks.

### Results

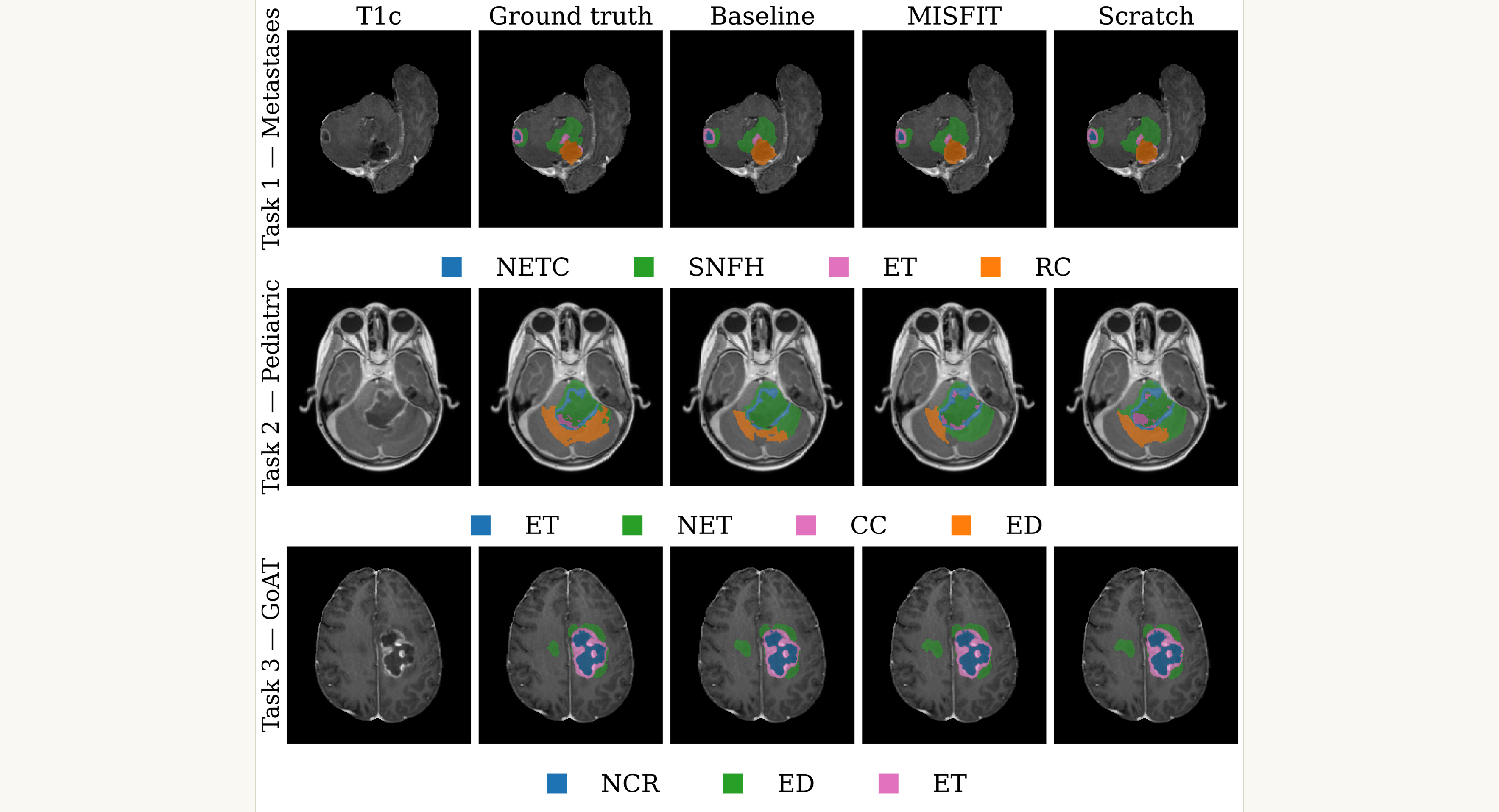
| Table 2 · BraTS-style average rank (lower is better), two-sided Wilcoxon p-values |              |              |                             |              |              |                            |
|-----------------------------------------------------------------------------------|--------------|--------------|-----------------------------|--------------|--------------|----------------------------|
| 5-fold cross-validation                                                           |              |              | Held-out validation         |              |              |                            |
| TASK                                                                              | BASE         | MISFIT       | P                           | BASE         | MISFIT       | P                          |
| 1                                                                                 | 1.503        | <b>1.497</b> | 0.62                        | <b>1.427</b> | 1.573        | <b>1.5×10<sup>-4</sup></b> |
| 2                                                                                 | 1.528        | <b>1.472</b> | <b>0.026</b>                | 1.503        | <b>1.497</b> | 0.83                       |
| 3                                                                                 | 1.518        | <b>1.482</b> | <b>0.0072</b>               | 1.513        | <b>1.487</b> | 0.32                       |
| Task                                                                              | MISFIT       | Scratch      | p                           | MISFIT       | Scratch      | p                          |
| 1                                                                                 | <b>1.439</b> | 1.561        | <b>2.1×10<sup>-5</sup></b>  | <b>1.436</b> | 1.564        | <b>0.0024</b>              |
| 2                                                                                 | <b>1.457</b> | 1.543        | <b>0.035</b>                | <b>1.440</b> | 1.560        | <b>0.0053</b>              |
| 3                                                                                 | <b>1.416</b> | 1.584        | <b>1.9×10<sup>-13</sup></b> | <b>1.424</b> | 1.576        | <b>7.4×10<sup>-5</sup></b> |

Against the **matched control** (upper vs. lower block, same architecture/patch/budget), MISFIT pretraining wins on all three tasks under both protocols — the isolated pretraining effect is real. Against the **established baseline**, the picture reverses: cross-validation favors the pretrained pipeline on Tasks 2–3, but that advantage vanishes on held-out validation, and the baseline pulls significantly ahead on Task 1 held-out ( $p = 1.5\times 10^{-4}$ ).

**Why the CV signal doesn't replicate** — the MAE encoder was pretrained on the pooled training split, giving it label-free exposure to images that later formed each cross-validation fold. That asymmetry inflates CV scores for Tasks 2–3; held-out scoring, which excludes the pretraining pool entirely, is unaffected and is why we treat it as the decisive protocol.



**Figure 1.** Validation loss (Dice, excl. background) across training. The MISFIT-pretrained arm reaches within 3.5–7.3% of the baseline's fully converged floor inside its 1,000-epoch budget — in far fewer epochs than the baseline needs (means: 3,107 / 3,908 / 4,252). The from-scratch SwinUNETR fails to converge at all on Task 2.



**Figure 2.** One representative training-cohort case per task: T1c, ground truth, and all three arms. All three agree closely on Tasks 1 and 3; on Task 2 the baseline recovers almost none of the cystic component (pink).



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## Explore MIST & MISFIT

Scan for code, documentation, and releases at [github.com/mist-medical](https://github.com/mist-medical)

### References

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